

NEWTON 8 W, GA, USA

WMO: 740050

Lat: 31.3127N

Lon: 84.4706W

Elev: 176

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 63828

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.3	27.5	13.1	10.8	37.1	17.7	13.5	37.6	8.2	67.0	6.9	61.2	0.4	N/A	0.300

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	94.9	74.7	93.0	74.9	91.2	74.9	78.9	88.3	78.1	87.5	77.4	86.6	3.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.7	139.9	82.4	75.7	135.3	81.7	75.0	132.1	81.1	42.5	88.4	41.6	87.8	40.8	86.6	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.8	6.0	5.0	DB	18.7	96.8	3.0	2.8	16.5	98.8	14.8	100.4	13.1	102.0	11.0	104.0	
			WB	18.0	80.5	3.6	0.7	15.5	81.0	13.4	81.5	11.4	81.9	8.8	82.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.0	48.8	52.5	59.0	65.2	73.0	79.1	80.6	80.3	76.4	67.1	56.7	51.9
	DBStd	13.32	10.11	8.99	8.22	6.31	5.33	3.10	2.49	2.69	4.24	7.40	8.37	9.39
	HDD50	366	150	73	21	1	0	0	0	0	0	1	29	91
	HDD65	1923	507	359	218	74	10	0	0	0	1	66	272	416
	CDD50	6189	114	142	298	457	714	873	947	939	792	531	231	151
	CDD65	2271	6	8	31	79	259	423	482	474	343	131	24	11
	CDH74	20664	17	60	309	872	2538	4008	4484	4176	2862	1119	177	42
	CDH80	8567	0	4	42	190	993	1869	2083	1866	1190	311	19	0
Wind	WSAvg	1.9	2.1	2.2	2.4	2.1	2.0	1.8	1.7	1.6	1.6	1.6	1.7	1.9
Precipitation	PrecAvg	49.6	4.8	4.0	5.6	3.1	2.3	5.4	6.0	4.8	4.7	2.8	3.1	3.5
	PrecMax	69.0	14.9	6.7	11.2	7.4	6.1	9.8	9.8	9.3	14.4	8.6	9.0	6.5
	PrecMin	32.7	0.9	0.6	0.4	0.6	0.1	1.5	3.6	2.2	0.8	0.4	0.4	0.9
	PrecStd	9.4	3.0	1.9	3.5	1.7	1.5	2.6	2.0	2.0	3.4	1.9	2.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.3	79.6	84.0	87.5	93.7	97.4	96.6	96.2	94.9	89.7	82.5	77.9
		MCWB	65.5	66.8	64.6	69.0	70.4	72.5	76.7	74.6	73.5	72.0	68.2	67.6
	2%	DB	73.0	75.6	80.8	84.8	90.7	94.8	94.0	93.6	91.5	86.3	78.9	74.9
		MCWB	63.5	63.8	64.3	67.3	70.4	74.4	76.1	75.9	73.3	68.3	66.1	65.6
	5%	DB	69.3	71.6	77.6	82.1	88.2	92.4	92.1	91.5	89.3	83.7	75.4	71.3
		MCWB	62.7	63.3	63.4	65.9	69.8	74.9	76.1	76.2	73.2	67.3	64.5	65.8
	10%	DB	65.8	68.2	74.4	79.6	86.0	89.8	90.1	89.4	87.2	80.7	72.2	68.0
		MCWB	60.5	61.0	61.9	64.5	69.5	74.7	76.0	76.0	73.1	66.7	63.1	63.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.1	69.1	69.3	72.6	75.3	79.4	79.7	80.3	78.2	76.2	73.1	71.2
		MCDB	72.7	75.6	76.1	80.7	83.1	90.0	89.2	89.3	86.9	83.0	77.4	75.7
	2%	WB	66.2	67.5	67.8	70.9	74.0	77.9	78.6	79.2	77.0	74.3	70.7	69.1
		MCDB	70.4	73.6	74.6	79.7	82.7	88.3	88.3	88.0	85.5	80.5	74.6	72.5
	5%	WB	64.3	65.6	66.2	69.5	73.0	77.0	78.0	78.2	76.1	72.4	68.3	67.2
		MCDB	68.3	71.8	73.7	77.2	82.2	87.0	87.7	87.0	84.9	79.0	72.8	70.8
	10%	WB	61.7	63.2	64.4	68.0	71.9	76.1	77.3	77.4	75.1	70.5	65.2	64.8
		MCDB	66.1	68.5	72.4	75.8	81.3	85.4	86.8	86.0	83.7	77.4	71.5	68.5
Mean Daily Temperature Range	5% DB	MDBR	23.0	23.4	24.4	24.9	23.5	20.4	19.3	18.4	20.3	24.0	25.7	22.6
		MCDBR	23.5	25.8	26.7	27.2	25.8	24.9	22.0	21.5	23.5	26.1	26.5	22.8
		MCWBR	14.5	14.6	12.4	11.5	9.1	7.7	7.1	7.6	8.9	12.0	14.7	14.7
	5% WB	MCDBR	19.0	20.8	21.5	21.5	20.5	19.9	19.2	18.8	19.8	19.7	19.1	18.1
		MCWBR	13.6	14.0	11.6	11.3	8.3	7.9	7.5	7.5	8.0	10.0	11.8	13.5
Clear-Sky Solar Irradiance	taub		0.326	0.333	0.355	0.384	0.433	0.479	0.508	0.497	0.425	0.374	0.351	0.332
	taud		2.511	2.487	2.441	2.359	2.255	2.188	2.118	2.147	2.334	2.450	2.456	2.512
	Ebn at Noon		286	295	294	288	273	259	251	252	268	276	274	278
	Edh at Noon		28	31	35	39	44	47	50	48	38	32	29	26
All-Sky Solar Radiation	RadAvg		961	1140	1518	1866	2047	1933	1914	1753	1578	1366	1076	836
	RadStd		60	134	125	148	142	141	116	119	129	135	85	80

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.40	N/A	N/A	N/A	N/A	N/A	N/A	-310	+416	N/A
(47)	Regional (38 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+288	+184

Nomenclature: See separate page